

IMO(29th) – 1988

First Day

1. Consider two concentric circles of radii $R, r (R > r)$ with center O . Fix P on the small circle and consider the variable chord PA of the small circle. Point B and C lie on the large circle; B, P, C are collinear and BC is perpendicular to AP .
 - (a) For which value(s) of $\angle OPA$ is the sum of $BC^2 + CA^2 + AB^2$ extremal?
 - (b) What are possible positions of the midpoints U of BA and V of AC as $\angle OPA$ varies?
2. Let n be an even positive integers. Let $A_1, A_2, \dots, A_{n+1}$ be sets having n elements each such that any two of them have exactly one element in common, while every element of their union belongs to at least two of the given sets. For which n can one assign to every element of the union one of the numbers 0 and 1 in such a manner that each of these sets has exactly $n/2$ zeroes?
3. A function f defined on the positive integers (and taking positive integers values) is given by:

$$f(1)=1, \quad f(3)=3, \quad f(2n)=f(n),$$

$$f(4n+1)=2f(2n+1)-f(n), \quad f(4n+3)=3f(2n+1)-2f(n)$$
 for all positive integers n . Determine with proof the number of positive integers less than or equal to 1988 for which $f(n)=n$.

Second Day

4. Show that the solution set of the inequality $\sum_{k=1}^{70} \frac{k}{x-k} \geq \frac{5}{4}$ is the union of disjoint half-open intervals with the sum of lengths 1988.
5. In a right angled triangle ABC let AD be the altitude drawn to the hypotenuse and let the straight line joining the incentres of the triangle ABD, ACD intersect the sides AB, AC at the points K, L respectively. If E and E_1 denote the areas of the triangles ABC and AKL respectively, show that $\frac{E}{E_1} \geq 2$.
6. Let a and b be two positive integers such that $ab+1$ divides a^2+b^2 . Show that $\frac{a^2+b^2}{ab+1}$ is a perfect square.

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